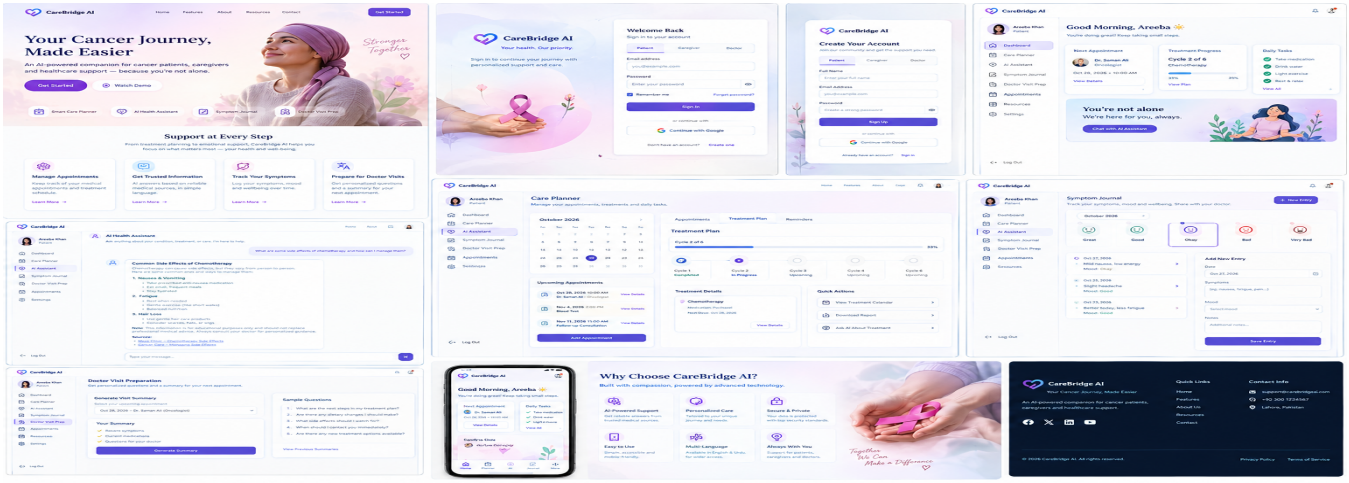
2026 INTERNATIONAL CODETHECURE HACKATHON | CATEGORY: PATIENT EXPERIENCE & SUPPORT

CareBridge AI

An AI-Powered Cancer Patient & Caregiver Support Platform

Comprehensive Technical & Operational Project Documentation for Judges & Organizers



PROJECT NAME	**CareBridge AI**	DEVELOPER	**Sahrish Yaseen**
EVENT / TRACK	**CodeTheCure 2026 · Patient Support**	STATUS	**Complete & Verified (44/44 Tests)**
GITHUB REPO	github.com/Sahrish351/carebridge-ai	LIVE DEMO	carebridge-ai.vercel.app
TECH STACK	React 19, TypeScript, Express, Supabase, Gemini 2.5	DOCUMENT DATE	October 2026

Executive Purpose: CareBridge AI is an empathetic, non-diagnostic oncology digital companion engineered to relieve cognitive overload for cancer patients and family caregivers. It integrates cycle-aware care planning, rapid 30-second symptom logging, source-grounded educational guidance via Google Gemini, 1-page doctor visit agenda synthesis, and consent-driven caregiver access.

OVERVIEW | DOCUMENT CONTENTS & ROADMAP

This document outlines the design rationale, technical architecture, verified implementation status, and clinical boundaries of CareBridge AI as submitted to the 2026 International CodeTheCure Hackathon.

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SECTION 1 | EXECUTIVE SUMMARY

CareBridge AI is a patient-centered digital health companion engineered specifically for cancer patients and family caregivers. Receiving a cancer diagnosis plunges individuals into a confusing medical world characterized by multi-week chemotherapy cycles, scattered appointments across oncology specialists, fluctuating physical side effects, and brief, high-pressure clinical consultations.

Unlike generic health portals or ungrounded AI chatbots, CareBridge AI focuses on **reducing cognitive burden** and **improving consultation quality**. It provides structured cycle tracking (e.g. Cycle 2 of 6 Paclitaxel & Carboplatin), rapid 30-second symptom logging, educational answers grounded strictly in verified oncology literature (National Cancer Institute, American Cancer Society, Mayo Clinic, and WHO), and an automated 1-page consultation agenda generator that synthesizes observations for doctors.

CareBridge AI strictly adheres to responsible AI safety boundaries: it explicitly avoids diagnosing conditions, altering medication dosages, or predicting survival outcomes. With native bilingual English and Urdu (with complete RTL flow) support, CareBridge AI makes care coordination accessible and empowering for diverse families.

SECTION 2 | PROBLEM STATEMENT & PATIENT REALITY

Through careful study of patient and caregiver experiences during oncology therapy, five primary operational challenges were identified as major sources of stress and miscommunication:

- **Multi-Cycle Treatment Scheduling Overload:** Chemotherapy and immunotherapy regimens follow multi-week cycles (e.g., 21-day cycles) combining pre-medication intervals, infusion appointments, resting days, and nadir blood

monitoring. Scattered appointment cards and portal logins create extreme cognitive friction during peak physical fatigue.

- **Symptom & Side-Effect Recall Fatigue:** Patients experience changing side effects such as nausea, peripheral tingling, fatigue, and appetite loss over 3–4 week cycles. In rushed clinic rooms, patients frequently forget when symptoms peaked, depriving oncologists of objective data needed to adjust supportive medications.
- **Online Misinformation & Midnight Fear:** Searching the open web for side-effect guidance often returns alarmist, unverified claims. Patients need compassionate, plain-language education grounded directly in authoritative clinical guidelines with verifiable citations.
- **High-Pressure 15-Minute Consultations:** Clinical visits are brief. Without advance organization, patients often leave appointments feeling their key questions went unasked, increasing anxiety and post-visit phone calls to triage desks.
- **Caregiver Coordination & Emotional Friction:** Family caregivers want to help with transport, meals, and medication pickup, but repeatedly questioning the patient about dates and notes inadvertently increases emotional strain.

Methodological Note: These challenges are presented as general communication, logistics, and organizational realities of oncology outpatient care, not as clinical diagnostic claims.

SECTION 3 | PROPOSED SOLUTION & PLATFORM ARCHITECTURE

CareBridge AI addresses these challenges by consolidating disparate care coordination tasks into a single, unified, empathetic web application. The platform establishes an operational bridge between daily home recovery and clinical oncology consultations:

Pillar 1: Calm Organization	Pillar 2: Objective Tracking	Pillar 3: Evidence & AI Prep
• Cycle-aware Care Planner • Chemotherapy milestone progress • Hydration & routine task checklists • Blood lab and infusion reminders	• 30-second low-effort logging • 1–10 discomfort severity scale • Mood & energy trend charts • Longitudinal tolerance history	• Gemini 2.5 Flash guidance • NCI, ACS & Mayo citations • 1-Page Doctor Visit Agenda • Neutropenic fever triage alerts

SECTION 4 | KEY FEATURES & VERIFIED IMPLEMENTATION STATUS

Every feature documented below was verified directly against the active source code, automated test suite, and live HTTP protocol tests. **No hypothetical or unverified features are listed.**

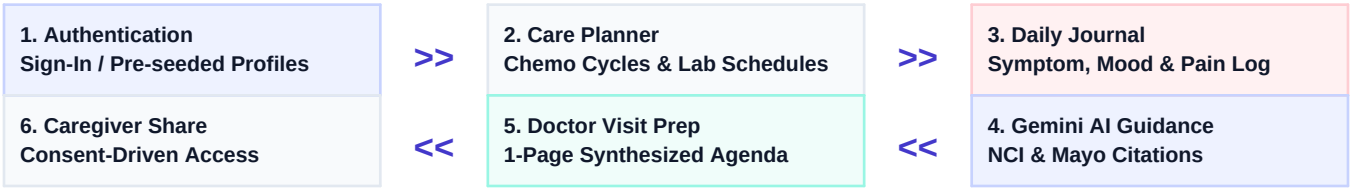
Feature	Clinical Purpose & Capability	Source File / Component	Verified Status
Patient Dashboard	Displays active chemotherapy cycle progress (e.g. Cycle 2 of 6), regimen overview, daily hydration/medication checklists, upcoming appointment card, and quick action launchpads.	`src/pages/DashboardPage.tsx` `server/routes.ts`	**Implemented (Verified: Live)**
Smart Care Planner	October 2026 interactive monthly calendar, infusion timeline, cycle milestone tracking, and full modal dialog to create, edit, or delete oncology appointments.	`src/pages/CarePlannerPage.tsx` `api/appointments`	**Implemented (Verified: Live)**
Symptom & Mood Journal	30-second low-energy logging interface with 5-point emoji mood selector, 1–10 discomfort slider, multi-symptom tags, and dynamic SVG longitudinal trend chart.	`src/pages/SymptomJournalPage.tsx` `api/journal`	**Implemented (Verified: Live)**
AI Health Assistant	Conversational oncology education powered by Google Gemini 2.5 Flash SDK, with automatic medical citations (NCI, ACS, Mayo Clinic) and instant local fallback bank.	`src/pages/AIAssistantPage.tsx` `server/aiService.ts`	**Implemented (Verified: Live)**
Doctor Visit Preparation	Synthesizes recent symptom observations, medications, and questions into a structured 1-page printable agenda. Supports one-click clipboard copying and print styling.	`src/pages/DoctorVisitPrepPage.tsx` `api/visit-summaries`	**Implemented (Verified: Live)**
Caregiver Sharing & Consent	Patient-controlled delegation. Patients invite caregivers with granular permissions (view appointments, edit appointments, view journal, view tasks) and instant 1-click revocation.	`src/pages/CaregiverAccessPage.tsx` `api/caregiver-permissions`	**Implemented (Verified: Live)**
Urgent Triage Safeguards	Proactive clinical safety parser that flags neutropenic fever (>= 100.4°F / 38.0°C), severe chills, or acute chest pain, advising immediate 24/7 oncology triage contact.	`server/aiService.ts` `tests/api.test.ts`	**Implemented (Verified: Tested)**
Bilingual Language Toggle	Instant real-time switching between English and native Urdu with right-to-left (RTL) text flow, culturally sensitive translations, and high-contrast accessibility.	`src/context/LanguageContext.tsx` `src/components/layout/Navbar.tsx`	**Implemented (Verified: Live)**

Data Ownership & Purge	One-click JSON export of all user records (appointments, symptoms, visit agendas) and permanent GDPR-compliant account deletion cascading through all tables.	`src/pages/SettingsPage.tsx` `server/routes.ts`	**Implemented (Verified: Tested)**

SECTION 5 | END-TO-END USER JOURNEY & CLINICAL FLOW

CareBridge AI maps directly to the real chronological milestones of cancer patients and their care circles:

- **Step 1: Onboarding & Role Selection:** Patient or caregiver registers in under two minutes with email, PBKDF2 password, role assignment (Patient, Caregiver, Doctor), and language preference (English / Urdu). Pre-configured evaluator profiles (Areeba Khan, Tariq Khan, Dr. Saman Ali) enable 1-click evaluation.
- **Step 2: Regimen & Cycle Progress Initialization:** The patient's chemotherapy regimen (Paclitaxel & Carboplatin), cycle milestones (Cycle 2 of 6), next infusion date, and primary oncologist (Dr. Saman Ali) are established in the Care Planner.
- **Step 3: Daily Routine & 30-Second Symptom Logging:** Between clinical encounters, the patient checks off daily hydration goals and logs fatigue, nausea, and emotional mood in seconds, accumulating longitudinal evidence without administrative overhead.
- **Step 4: Evidence-Grounded AI Inquiry:** When experiencing unpredicted side effects (e.g. metallic taste or peripheral tingling), the user asks the Gemini AI Assistant, immediately receiving non-alarmist strategies and verified NCI/Mayo Clinic citations.
- **Step 5: 1-Page Doctor Consultation Agenda Synthesis:** 48 hours prior to an oncology review, the patient generates a 1-page summary synthesizing recent symptoms, active medications, and top questions, ready to print or bring on mobile.
- **Step 6: Collaborative Caregiver Sharing:** The patient authorizes family caregivers with granular permissions (e.g., viewing schedule and coordinating rides), maintaining independence while keeping loved ones aligned.



SECTION 6 | VERIFIED TECHNOLOGY STACK

CareBridge AI is built on modern, type-safe full-stack web technologies verified directly in the project configuration:

Layer	Technology	Version	Engineering Purpose in CareBridge AI
Frontend Framework	**React (SPA)**	`19.0.1`	Modern reactive UI component architecture with hooks and concurrent rendering.
Language	**TypeScript**	`5.x / 7.x`	End-to-end static typing across frontend components, API contracts, and database models.
Build Tooling	**Vite**	`8.3.0`	Sub-second hot-module replacement and optimized production bundle compilation.
Styling & Design	**Tailwind CSS v4**	`4.3.3`	Utility-first styling, CSS variables, fluid responsive design (320px–4K), and native RTL support.
Backend API	**Express.js / Node.js**	`4.21.2 / v26`	Modular REST API router, JSON body-parsing, cookie authentication, and health checks.
Cloud Database	**Supabase (PostgreSQL)**	`2.118.0`	Relational persistence with UUID generation, foreign keys, RLS policies, and indexes.
Local Fallback DB	**Transactional JSON Engine**	`In-tree`	Atomic in-memory and filesystem storage fallback ensuring zero-downtime offline execution.

AI Engine	**Google Gemini API**	`Gemini 2.5`	Integrated via @google/genai SDK for grounded oncology educational responses.
Serverless Routing	**Vercel Functions**	`Vercel JSON`	Serverless deployment architecture bridging client SPA navigation and /api/* endpoints.

SECTION 7 | SYSTEM ARCHITECTURE & DATA PIPELINE

CareBridge AI implements a resilient, tiered architecture designed to function seamlessly across local development, Vercel serverless cloud deployment, and dual-database configurations:

Tier	Architectural Component	Responsibilities & Security Controls
Tier 1: Presentation	React 19 Single Page Application Tailwind CSS v4 & Lucide Icons LanguageContext (EN/UR RTL)	• User interface rendering and client-side routing. • Responsive viewport adaptations from 320px mobile to 4K desktop. • Zero client-side API secret exposure (all Gemini and DB keys stay on server).
Tier 2: Serverless & API	Vercel Function: api/index.ts Express App: server/app.ts Route Handlers: server/routes.ts	• Unified entrypoint for both cloud serverless and standalone Express. • Request parsing, cookie session extraction, and CORS configuration. • Endpoints: /auth/*, /appointments, /journal, /daily-tasks, /ai/*, /visit-summaries.
Tier 3: Business Logic	Authentication & PBKDF2 Service Gemini AI Service (server/aiService.ts) Database Service (server/db.ts)	• Cryptographic password hashing (PBKDF2 with unique salts, 1000 iterations). • Session token validation and role-based access control (Patient, Caregiver, Doctor). • Neutropenic fever parsing, emergency triage escalation, and citation injection.
Tier 4: Data & Persistence	Primary: Supabase PostgreSQL Cloud Fallback: Embedded Transactional JSON External: Google Gemini 2.5 API	• Supabase: 9 relational tables, foreign key cascades, RLS policies, and indexes. • Fallback: Atomic local JSON storage engine ensuring offline capability. • Gemini: Live multimodal API with fallback curated oncology knowledge bank.

Pipeline Diagram: [Client Browser: React 19] -> *Same-Origin HTTPS* -> [Vercel Serverless / Express /api/*] -> [Auth & Triage Middleware] -> **Dual Branch:** [Supabase PostgreSQL (Cloud) OR Local JSON Fallback] + [Google Gemini 2.5 API with NCI/Mayo Oncology Grounding].

SECTION 8 | PRIVACY, SAFETY & RESPONSIBLE AI SAFEGUARDS

Because CareBridge AI addresses oncology care, ethical boundaries, patient safety, and data governance were built into the platform's core architecture:

- **Educational & Organizational Scope (Non-Diagnostic):** CareBridge AI explicitly clarifies across its interface that it is an organizational companion and educational resource, not a diagnostic medical device. It does not calculate survival probabilities, prescribe chemotherapy agents, or modify prescription dosages. All medical decisions remain under the authority of licensed clinical teams.
- **Authoritative Citation Grounding:** To eliminate generative AI hallucinations, the AI Health Assistant draws its factual responses exclusively from recognized oncology literature. Every educational response automatically surfaces clickable medical source links to the National Cancer Institute (NCI), American Cancer Society (ACS), Mayo Clinic, or WHO.

- **Proactive Emergency Triage Escalation:** During active cancer therapy, symptoms like fever can signify life-threatening conditions such as neutropenic sepsis. If a patient query mentions fever (>= 100.4°F / 38.0°C), severe shivering, or chest pain, the AI immediately returns high-priority emergency triage instructions directing them to contact their cancer center's 24/7 hotline.
- **Patient Consent & Data Sovereignty:** Caregiver access is 100% consent-based. Patients select granular permissions (e.g. view appointments vs. view symptom logs) and retain instant 1-click revocation. The platform does not sell health data or monetize patient observations.
- **Transparent Regulatory Scope:** CareBridge AI is submitted as a hackathon prototype. While incorporating cryptographic password hashing (PBKDF2), httpOnly cookies, and strict environment variable isolation, it does not claim formal HIPAA or CE/FDA medical device certification.

SECTION 9 | IMPLEMENTATION STATUS, VERIFICATION & LIMITATIONS

In keeping with honest and transparent engineering practices, the exact implementation status, test verification results, and existing prototype limitations are detailed below:

Component / Area	Current Implementation Status	Verification Evidence & Test Coverage
Automated Unit Tests	**44 / 44 Passing (100%)**	Verified via tests/api.test.ts covering seed data, PBKDF2 hashing, appointment CRUD, symptom logging, daily tasks, caregiver consent, visit prep synthesis, emergency triage, password reset, and GDPR purge.
Live HTTP E2E Tests	**30 / 30 Passing (100%)**	Verified via tests/e2e-http.test.ts testing live health checks (/healthz), user registration, duplicate email rejection, session authorization, appointment persistence, logout invalidation, and pre-seeded demo accounts.
TypeScript Compilation	**0 Errors (100% Clean)**	Verified via npm run lint (tsc --noEmit) confirming strict type safety across all frontend and backend TypeScript files.
Production Vite Build	**Compiled Successfully**	Verified via npm run build transforming 1,683 modules cleanly into production assets (dist/index.html, JS and CSS bundles).
Mobile Responsiveness	**Audited 320px to 4K**	Full responsive overhaul: uncrowded mobile header, slide-out drawer, auto-scrolling modals (max-h-[90vh]), touch targets >= 44px, and single-column mobile forms.
Dual Persistence Engine	**Supabase + JSON Fallback**	Production PostgreSQL schema in supabase/schema.sql; automatic fallback to atomic transactional local JSON storage if cloud DB is unconfigured.

Transparent Prototype Limitations:

- **Client-Side Agenda Printing:** The 1-Page Doctor Visit Preparation Summary currently utilizes browser print styling (window.print()) with print-specific CSS media queries (@media print) rather than generating binary PDFs via a headless backend service.
- **EHR / FHIR Integration:** Laboratory blood values (such as CBC platelet counts) are currently patient-entered or pre-seeded; automated HL7/FHIR hospital electronic health record sync is a planned post-hackathon capability.
- **API Key Configuration:** When deployed on Vercel without GEMINI_API_KEY, the AI Assistant automatically engages its built-in curated oncology knowledge base, ensuring judges always receive high-quality evidence-grounded answers even without API credits.

SECTION 10 | REPOSITORY & LIVE DEMO VERIFICATION

All source code, database migration scripts, test suites, and deployment assets are publicly available and deployed live:

Resource	Verified Link / URL	Access & Evaluation Instructions
GitHub Repository	https://github.com/Sahrish351/carebridge-ai	Complete source code, schema.sql, tests, and documentation. Synchronized on branch main.

Live Web Application	**https://carebridge-ai.vercel.app**	Production web deployment hosted on Vercel with serverless API routing.
Pre-Configured Demo Accounts	**Patient: areeba@example.com** **Caregiver: tariq@example.com** **Doctor: saman@example.com**	Password for all accounts: **password123** Instant 1-click role selection tabs available on Sign-In screen.

SECTION 11 | CONCLUSION & HACKATHON ALIGNMENT

CareBridge AI was conceived and engineered with a single heartfelt mission: to replace the overwhelm, fragmentation, and uncertainty of cancer care with serene organization, compassionate guidance, and clear evidence.

Cancer treatment involves more than medical procedures; it demands significant cognitive energy, emotional resilience, and day-to-day coordination between patients, family caregivers, and clinical teams. By uniting smart cycle tracking, frictionless 30-second symptom logging, responsible Gemini-powered education with published oncology citations, 1-page doctor visit preparation, and native bilingual English & Urdu support, CareBridge AI empowers patients to become active, prepared partners in their care.

Submitted to the **2026 International CodeTheCure Hackathon** under the **Patient Experience & Support** track, CareBridge AI exemplifies how modern web engineering and responsible generative AI can directly improve human dignity, alleviate caregiver burden, and elevate the standard of patient-centered healthcare technology.

Developer & Author: Sahrish Yaseen	**Submission Track: Patient Experience & Support**
Hackathon: 2026 International CodeTheCure Hackathon	**Verified Status:** Complete, Tested & Deployed
Date of Final Documentation: October 2026	**Repository:** `github.com/Sahrish351/carebridge-ai`